# Sieve Analysis of Lead–Zinc Ore

**UMT302L Materials Processing Laboratory**

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**Sample Assigned:** Lead–Zinc Ore (Powdered)
**Apparatus:** Vibratory Sieve Shaker
**Settings:** Amplitude $A = 1.1\,\text{mm}$, Time $t = 10\,\text{min}$

## 1 Objective

- Obtain the particle size distribution for a comminuted Lead–Zinc ore sample using a vibratory sieve shaker.
- Plot the direct size distribution curve (% weight retained vs. sieve size fraction).
- Plot the cumulative distribution curves (cumulative oversize and cumulative undersize vs. sieve size).
- Calculate the Fineness Number and determine characteristic cut sizes ($D_{10}, D_{50}, D_{80}$) of the ore powder.
- Evaluate the degree of liberation for downstream froth flotation and verify reproducibility across two trials ($T_1$ and $T_2$).

## 2 Theory

### 2.1 Comminution and Mineral Liberation

Run-of-mine (ROM) lead–zinc ores consist of valuable sulfide minerals (galena PbS and sphalerite ZnS) locked inside a barren gangue matrix (primarily quartz and carbonates). Physical separation techniques like froth flotation cannot separate composite grains; the valuable minerals must first be exposed as distinct, free particles. Therefore, **comminution** (crushing and grinding) is universally the mandatory *first step* in processing an ore. Comminution breaks rock along natural grain boundaries and internal flaws, reducing particle size to achieve mineral liberation.

The **liberation size** is defined as the particle diameter to which an ore must be ground so that the valuable mineral grains exist predominantly as discrete, unbound phases rather than composite middlings. Achieving liberation involves a balance:

- **Undergrinding (particles coarser than liberation size):** Locked composite particles float poorly or float unselectively, causing metal loss in the tailings or lowering concentrate grade.
- **Overgrinding (excessive fines, $< 15 - 20\,\mu$m):** Grinding consumes significant electrical energy. Producing excessive "slimes" harms flotation: fines have low momentum, collide inefficiently with air bubbles, and consume excess reagents.

Sieve analysis provides the particle size distribution needed to monitor grinding and verify that the product matches the liberation size window.

### 2.2 Sieve Arrangement and Standard Sieve Scales

Test sieves are arranged vertically in **strictly decreasing order of aperture opening from top to bottom**, resting on a solid collection pan. This arrangement is necessary because:

1. It allows stepwise classification: coarse particles are caught on top, protecting the finer, delicate lower wire meshes from heavy mass loading and damage.
2. It prevents mesh blinding: fine apertures would quickly clog if loaded with the full unclassified bulk charge.

3. It provides efficient separation under vibration, allowing undersize particles to pass progressively downward.

Standard test sieves follow a geometric progression. In the **British Standard Sieve (BSS 410)** and the **Tyler Standard Scale**, the aperture area doubles between consecutive principal sieves. Since screen area is proportional to the square of aperture width ($A = d^2$), the linear aperture ratio between consecutive sieves is:

$$\text{Aperture Ratio} = \sqrt{2} \approx 1.414 \tag{1}$$

For closer sizing, intermediate sieves with a ratio of $\sqrt[4]{2} \approx 1.189$ are used. The sieves used in our experiment ($250\,\mu\text{m}, 150\,\mu\text{m}, 105\,\mu\text{m}, 75\,\mu\text{m}, 53\,\mu\text{m}, 37\,\mu\text{m}$) closely adhere to this $\sqrt{2} \approx 1.41$ geometric series ($150/105 = 1.43$, $105/75 = 1.40$, $75/53 = 1.42$, $53/37 = 1.43$).

### 2.3 Operational Limits of Dry Sieve Analysis

In dry vibratory sieving, the practical lower size limit that can be separated is approximately **$37\,\mu$m to $38\,\mu$m** (400 mesh). Several physical factors limit dry sieving below this size:

1. **Surface forces dominate over gravity/inertia:** Gravitational and inertial forces scale with particle volume ($\propto d^3$), whereas interparticle surface forces (van der Waals attraction and electrostatic charges) scale with surface area or diameter ($\propto d$). Below $\sim 37\,\mu$m, surface forces dominate, causing fine particles to agglomerate into loose flocs or stick to the wire mesh and sieve walls rather than passing through openings.
2. **Aperture blinding:** Grains close to the aperture size wedge into the tiny openings. Fine wire cloth lacks the stiffness to dislodge trapped particles without damage.
3. **Air resistance and humidity:** Micro-particles have very low settling velocities and remain suspended in air currents inside the shaker. Ambient humidity can also form capillary moisture bridges between particles, causing caking.

For sub-sieve particles ($< 37\,\mu$m), wet sieving, sedimentation/decantation (governed by Stokes' law), or laser diffraction must be used instead.

## 3 Materials and Procedure

### 3.1 Materials Required

- Sieve set: $250\,\mu\text{m}, 150\,\mu\text{m}, 105\,\mu\text{m}, 75\,\mu\text{m}, 53\,\mu\text{m}, 37\,\mu\text{m}$, bottom pan, and lid.
- Vibratory sieve shaker.
- Comminuted Lead–Zinc ore sample.
- Electronic weighing balance ($\pm 0.01\,\text{g}$).
- Soft camel-hair brush and sampling trays.

### 3.2 Procedure

1. **Mixing and Sampling (Coning and Quartering):** The bulk powdered ore was mixed thoroughly. It was piled into a symmetrical cone, flattened into a uniform circular cake, and divided into four equal quadrants. Opposite diagonal quarters were discarded, and the remaining quarters were recombined. This was repeated to obtain an unbiased, representative analytical sample of $\approx 115 - 120\,\text{g}$.
2. **Stacking Sieves:** Cleaned sieves were stacked vertically in decreasing aperture order: Pan, $37\,\mu$m, $53\,\mu$m, $75\,\mu$m, $105\,\mu$m, $150\,\mu$m, and $250\,\mu$m.
3. **Loading Sample:** The weighed ore sample was loaded onto the top ($250\,\mu$m) sieve, and the lid was placed on top.
4. **Shaking:** The stack was clamped into the vibratory sieve shaker. The shaker was run at amplitude $A = 1.1\,\text{mm}$ for 10 minutes (per the lab manual protocol).

5. **Weighing:** The shaker was turned off and allowed to settle. Each sieve was opened separately, and the retained material was carefully transferred to tare paper using a soft brush on the underside of each mesh. Each fraction was weighed and recorded.
6. The experiment was carried out for two independent runs: Trial 1 ($T_1$) and Trial 2 ($T_2$).

**Observations:** Minimal airborne dust was observed during the run. The majority of the mass was retained on the intermediate sieves (75 $\mu$m and 53 $\mu$m). Some near-mesh grains wedged into the 75 $\mu$m sieve and were recovered by gentle reverse brushing. During Trial 2, the vibratory shaker showed slight amplitude fluctuations between 1.0 mm and 1.1 mm.

## 4 Results

**Initial sample weight loaded:** $\approx 120.0$ g nominal charge.
**Total recovered mass:** Trial 1 ($T_1$) = **117.32 g**, Trial 2 ($T_2$) = **115.84 g**.

The measured weights and calculated distributions for Trial 1 and Trial 2 are presented in Table **??**.

**Table 1:** Sieve analysis data and particle size distribution for Lead–Zinc ore ($T_1$ and $T_2$).

| Sl. No. | Sieve Size ($\mu$m) | Wt. Retained (g) | | Wt. Retained (%) | | Cum. Oversize (%) | | Cum. Undersize (%) | |
|---|---|---|---|---|---|---|---|---|---|
| | | $T_1$ | $T_2$ | $T_1$ | $T_2$ | $T_1$ | $T_2$ | $T_1$ | $T_2$ |
| 1 | +250 | 8.95 | 10.15 | 7.63 | 8.76 | 7.63 | 8.76 | 92.37 | 91.24 |
| 2 | $-250+150$ | 11.91 | 15.08 | 10.15 | 13.02 | 17.78 | 21.78 | 82.22 | 78.22 |
| 3 | $-150+105$ | 17.67 | 19.62 | 15.06 | 16.94 | 32.84 | 38.72 | 67.16 | 61.28 |
| 4 | $-105+75$ | 51.22 | 44.07 | 43.66 | 38.04 | 76.50 | 76.76 | 23.50 | 23.24 |
| 5 | $-75+53$ | 22.82 | 21.14 | 19.45 | 18.25 | 95.95 | 95.01 | 4.05 | 4.99 |
| 6 | $-53+37$ | 4.49 | 5.45 | 3.83 | 4.70 | 99.78 | 99.72 | 0.22 | 0.28 |
| 7 | $-37$ (Pan) | 0.26 | 0.33 | 0.22 | 0.28 | 100.00 | 100.00 | 0.00 | 0.00 |
| **Total Recovered:** | | **117.32** | **115.84** | **100.00** | **100.00** | **Mean Fineness Number = 5.078 ± 0.039** | | | |

## 5 Calculations

The calculations are performed following the formulas given in Section 6 of the lab manual:

1. **Weight % Retained:**

$$\text{Weight \% Retained} = \frac{\text{weight grams retained}}{\text{Total weight of sample}} \times 100 \tag{2}$$

*Sample calculation for fraction $-105+75\,\mu m$ (Trial 1):*

$$\text{Weight \%} = \frac{51.22\,\text{g}}{117.32\,\text{g}} \times 100 = \mathbf{43.66}\%$$

2. **Cumulative % Retained (Oversize):**

$$\text{Cumulative \% Retained} = \frac{\text{cumulative grams retained}}{\text{Total weight of sample}} \times 100 \tag{3}$$

*Sample calculation for cumulative oversize retained down to 75 $\mu m$ sieve (Trial 1):*

$$\text{Cum. Grams} = 8.95 + 11.91 + 17.67 + 51.22 = 89.75\,\text{g}$$

$$\text{Cumulative \% Retained} = \frac{89.75\,\text{g}}{117.32\,\text{g}} \times 100 = \mathbf{76.50}\%$$

3. **Cumulative % Retained (Undersize):**

$$\text{Cumulative \% Undersize} = 100 - \text{Cumulative \% Oversize} \tag{4}$$

$$\text{Cumulative \% Undersize for } 75\,\mu\text{m} = 100 - 76.50 = \mathbf{23.50}\%$$

4. **Fineness Number (Formula from Lab Manual):**

$$\text{Fineness Number} = \frac{\sum(\text{cumulative grams retained})}{100} \tag{5}$$

For Trial 1 ($T_1$):

$$\text{FN}_{T_1} = \frac{8.95 + 20.86 + 38.53 + 89.75 + 112.57 + 117.06 + 117.32}{100} = \frac{505.04}{100} = \mathbf{5.0504}$$

For Trial 2 ($T_2$):

$$\text{FN}_{T_2} = \frac{10.15 + 25.23 + 44.85 + 88.92 + 110.06 + 115.51 + 115.84}{100} = \frac{510.56}{100} = \mathbf{5.1056}$$

$$\text{Average Fineness Number} = \frac{5.0504 + 5.1056}{2} = \mathbf{5.078 \pm 0.039}$$

5. **Characteristic Particle Diameters (from Cumulative Undersize Curve):** Interpolated from the cumulative passing curve:
   - $D_{10}$ (10% passing): **58.6** $\mu$m
   - $D_{50}$ (Median particle diameter, 50% passing): **93.5** $\mu$m
   - $D_{80}$ (Grind size indicator, 80% passing): **151.6** $\mu$m

# 6 Plots and Graphs

## 6.1 Direct Plot: Weight Retained vs. Sieve Size

Figure 1a shows the direct plot of weight percentage retained across each sieve fraction. Both trials show a clear modal peak in the $-105 + 75\,\mu$m fraction (43.66% in $T_1$, 38.04% in $T_2$).

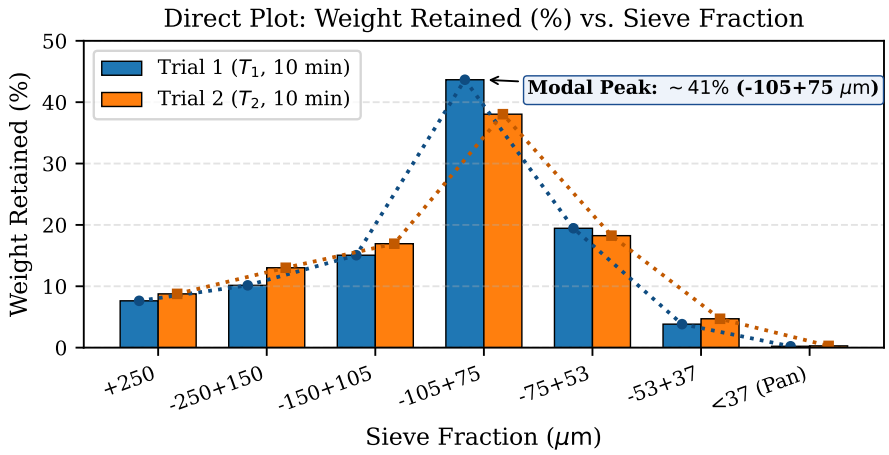



**Figure 1:** Direct plot: Weight retained (%) versus sieve fraction for Trial 1 ($T_1$) and Trial 2 ($T_2$).

## 6.2 Cumulative Plot: Cumulative Oversize and Undersize vs. Sieve Size

Figure 1b shows the cumulative oversize and undersize curves plotted against sieve aperture on a semi-log scale. The median diameter ($D_{50} = 93.5\,\mu$m) and 80% passing size ($D_{80} = 151.6\,\mu$m) are marked.

# 7 Discussion

- **Size Distribution Characteristics:** The crushed lead–zinc ore exhibits a unimodal distribution. Over 75.7% of the total sample mass is concentrated between 53 $\mu$m and 150 $\mu$m, with the modal fraction at $-105 + 75\,\mu$m accounting for an average of 40.85%.

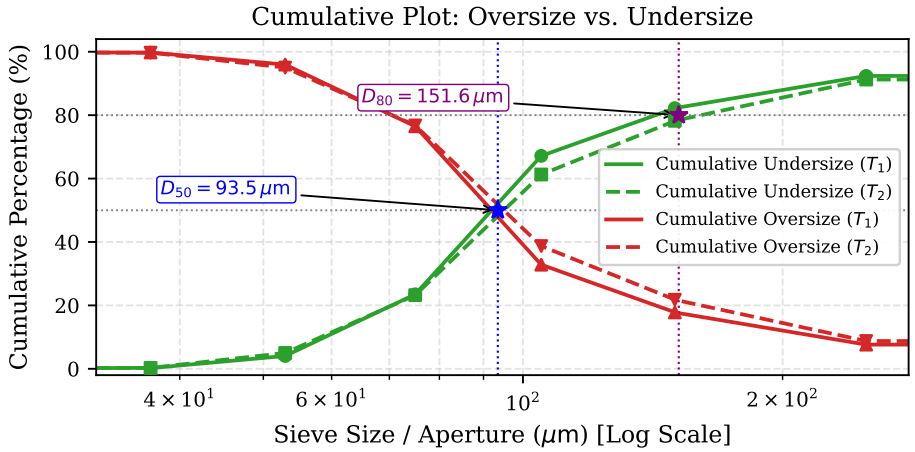



**Figure 2:** Cumulative plot: Cumulative oversize (%) and cumulative undersize (%) versus sieve aperture size.

- **Fines Generation:** The pan fraction ($< 37\,\mu$m) is very small (0.22% in $T_1$, 0.28% in $T_2$). This indicates that comminution did not produce excessive slimes, which is desirable because slimes float poorly and waste flotation reagents.
- **Liberation and Beneficiation Suitability:** For lead–zinc sulfide ores, the optimal size range for froth flotation of galena (PbS) and sphalerite (ZnS) is typically $20\,\mu$m to $100\,\mu$m. With a median size $D_{50} = 93.5\,\mu$m and 64.2% of the ore passing $105\,\mu$m, the majority of the material is close to liberation. The coarse $+105\,\mu$m fraction ($\approx 36\%$) consists of locked middlings that would be classified by a hydrocyclone in a commercial circuit and recycled back to the ball mill for regrinding.
- **Repeatability:** The results between Trial 1 and Trial 2 show close agreement. The cumulative oversize at the critical $75\,\mu$m screen was 76.50% in $T_1$ and 76.76% in $T_2$ (a difference of only 0.26%). The Fineness Numbers (5.05 vs. 5.11) differ by only 1.09%.
- **Sources of Error:**
  1. *Shaker Amplitude Fluctuations:* The nominal amplitude was set to 1.1 mm; however, during Trial 2, minor amplitude fluctuations between 1.0 mm and 1.1 mm were observed. A slightly lower amplitude reduces bed fluidization, which directly accounts for the slightly higher retained coarse mass on the $+250\,\mu$m and $+150\,\mu$m sieves in Trial 2.
  2. *Near-mesh blinding:* Particles close to the wire aperture size wedged in the $75\,\mu$m and $53\,\mu$m meshes, requiring gentle brush recovery.
  3. *Dust loss:* Minor dust loss during sample transfer and weighing accounted for a small mass deficit ($\approx 2 - 4$ g from the initial charge).
  4. *Particle shape:* Galena cleaves into cubic shapes while silicate gangue forms irregular grains. Non-spherical grains pass through square mesh openings based on their narrowest dimension rather than volume, slightly shifting the apparent distribution.

## 8 Conclusions

1. Sieve analysis of the crushed Lead–Zinc ore sample was successfully carried out across sieves ranging from $250\,\mu$m to $37\,\mu$m.
2. The size distribution is unimodal, with the peak mass retained on the $-105 + 75\,\mu$m fraction ($40.85 \pm 3.97\%$).
3. The characteristic particle diameters obtained from the cumulative undersize curve are:

$$D_{10} = 58.6\,\mu\text{m}, \qquad D_{50} = 93.5\,\mu\text{m}, \qquad D_{80} = 151.6\,\mu\text{m}$$

4. The Fineness Number of the ore sample was determined to be $\mathbf{5.078 \pm 0.039}$.
5. The results between Trial 1 and Trial 2 showed high repeatability overall (agreement within 0.26% at the $75\,\mu$m screen). Minor differences in upper sieve fractions were primarily attributed to observable shaker amplitude fluctuations (1.0 to 1.1 mm) during the second run.

6. Approximately 64% of the ore is finer than 105 $\mu$m, which is suitable for mineral liberation in froth flotation, while the coarser fraction (+105 $\mu$m) should be recycled in closed-circuit milling.